Mathematics & Statistics | Lancaster University

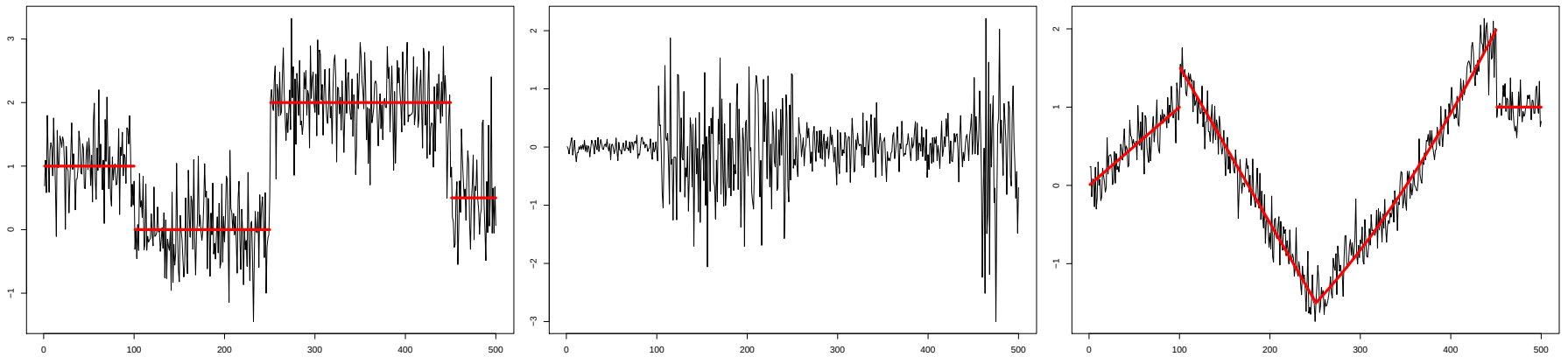
Methods

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Changepoints

For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

There are many different types of change.



Current state

Lots of work in the area in the last 10 years around:

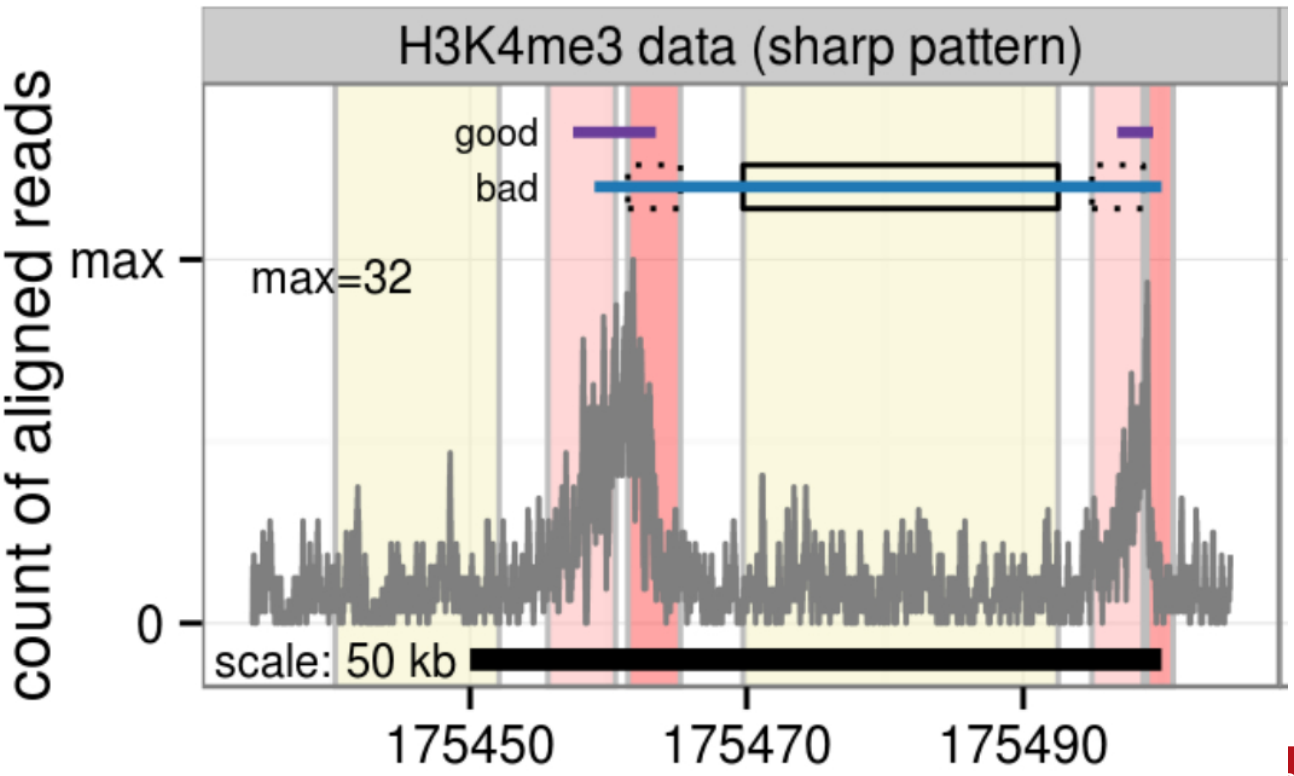
- Efficient univariate search methods for independent segments
- Still lots of work focussed on changes in mean
- First serious look at confidence intervals
- More non-parametric approaches appearing
- Field is starting to branch into multivariate changepoint problems

Environmental challenges

- Keeping some parameters constant whilst allowing others to change - but joint estimation
- Within and across season changes
- Dealing with correlation between series
- Lead/lag series effects
- Spatio-temporal changes
- Bridging process models and statistical models
- Real-time analysis for complex models
- Early warnings

Changepoints and ML

- Label some areas as containing suspected changepoint
- Label some areas as definitey not changepoint
- Use this to inform penalty choice - balancing false positive and false negative



Opportunities: cpts+ML

- Segment the data to get better inputs to ML algorithms
- Look for changepoints over ML output - real time

Opportunities

What do people think are the most important problems to tackle first?